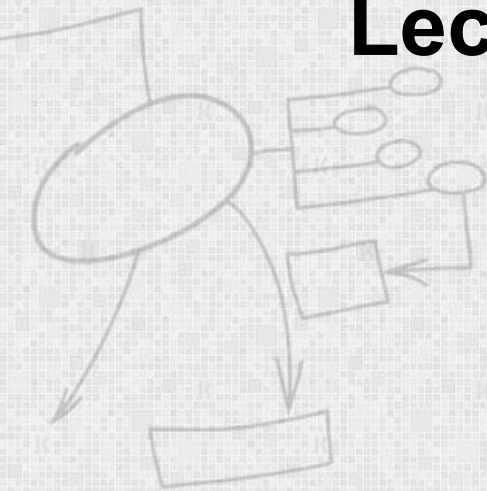


Database Engineering

Lecture 10: Data Warehousing

By

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Definitions

- **Data Warehouse**

- A subject-oriented, integrated, time-variant, non-updatable collection of data used in support of management decision-making processes

- ***Subject-oriented:*** e.g. sales, patients, customers etc
- ***Integrated:*** consistent naming conventions, formats, encoding structures; from multiple data sources
- ***Time-variant:*** can study trends and changes
- ***Non-updatable:*** read-only, periodically refreshed

- **Data Mart**

- A data warehouse that is limited in scope (Single subject, fewer sources)



Need for Data Warehousing

- Integrated, company-wide view of high-quality information (from disparate databases)
- Separation of *operational* and *informational* systems and data (for improved performance)

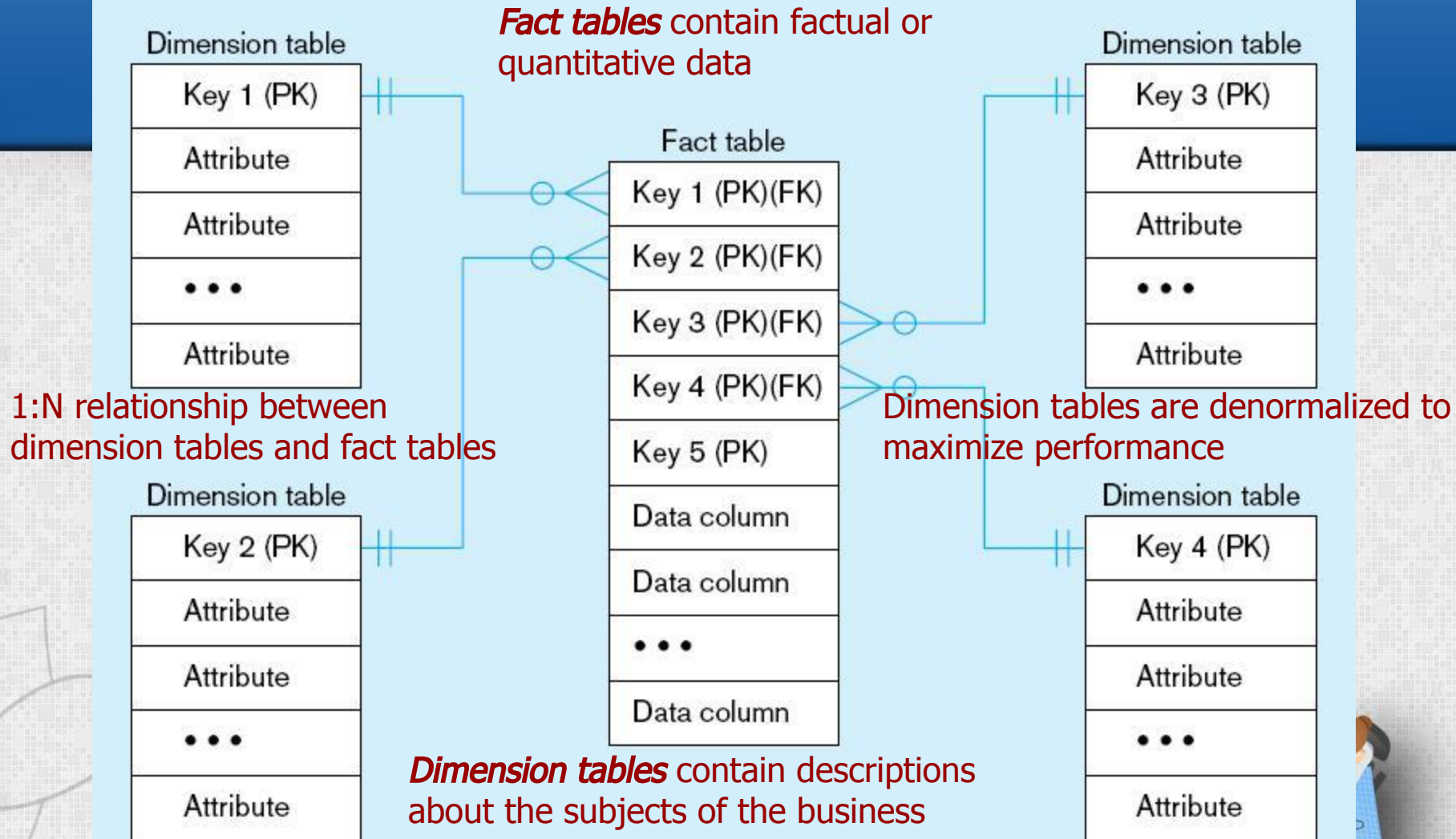


Extract, Transform, Load

- Retrieving data from different sources is extract
- Transform is selecting column, changing values, computing values, cleansing, missing value handling, encoding, aggregating and so on
- Load is inserting data into data mart



Components of a star schema



Excellent for ad-hoc queries, but bad for online transaction processing

Star schema example

Fact table provides statistics for sales broken down by product, period and store dimensions

PRODUCT

<u>Product Code</u>
Description
Color
Size

PERIOD

<u>Period Code</u>
Year
Quarter
Month
Day

SALES

<u>Product Code</u>
<u>Period Code</u>
<u>Store Code</u>
Units Sold
Dollars Sold
Dollars Cost

STORE

<u>Store Code</u>
Store Name
City
Telephone
Manager



Star schema with sample data

Product

<u>Product Code</u>	Description	Color	Size
100	Sweater	Blue	40
110	Shoes	Brown	10 1/2
125	Gloves	Tan	M
...			

Period

<u>Period Code</u>	Year	Quarter	Month
001	2010	1	4
002	2010	1	5
003	2010	1	6
...			

Sales

<u>Product Code</u>	<u>Period Code</u>	<u>Store Code</u>	Units Sold	Dollars Sold	Dollars Cost
110	002	S1	30	1500	1200
125	003	S2	50	1000	600
100	001	S1	40	1600	1000
110	002	S3	40	2000	1200
100	003	S2	30	1200	750
...					

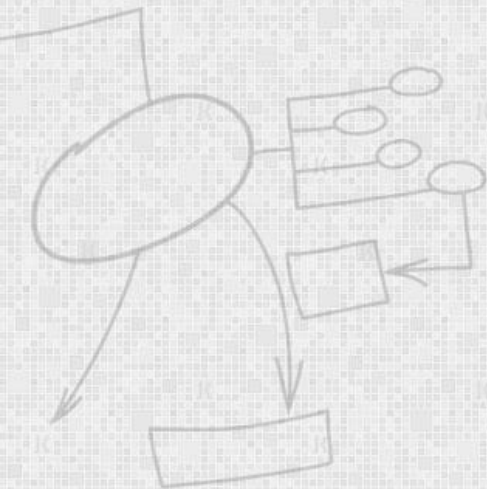
Store

<u>Store Code</u>	Store Name	City	Telephone	Manager
S1	Jan's	San Antonio	683-192-1400	Burgess
S2	Bill's	Portland	943-681-2135	Thomas
S3	Ed's	Boulder	417-196-8037	Perry
...				



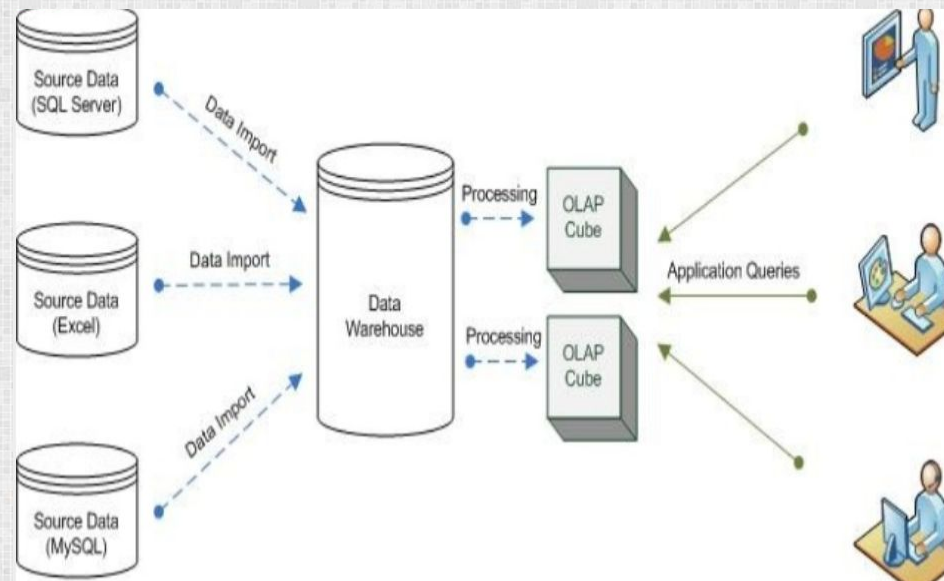
Grain of the Fact Table

- Granularity of Fact Table—what level of detail do you want?
 - Transactional grain—finest level
 - Aggregated grain—more summarized
 - Finer grains

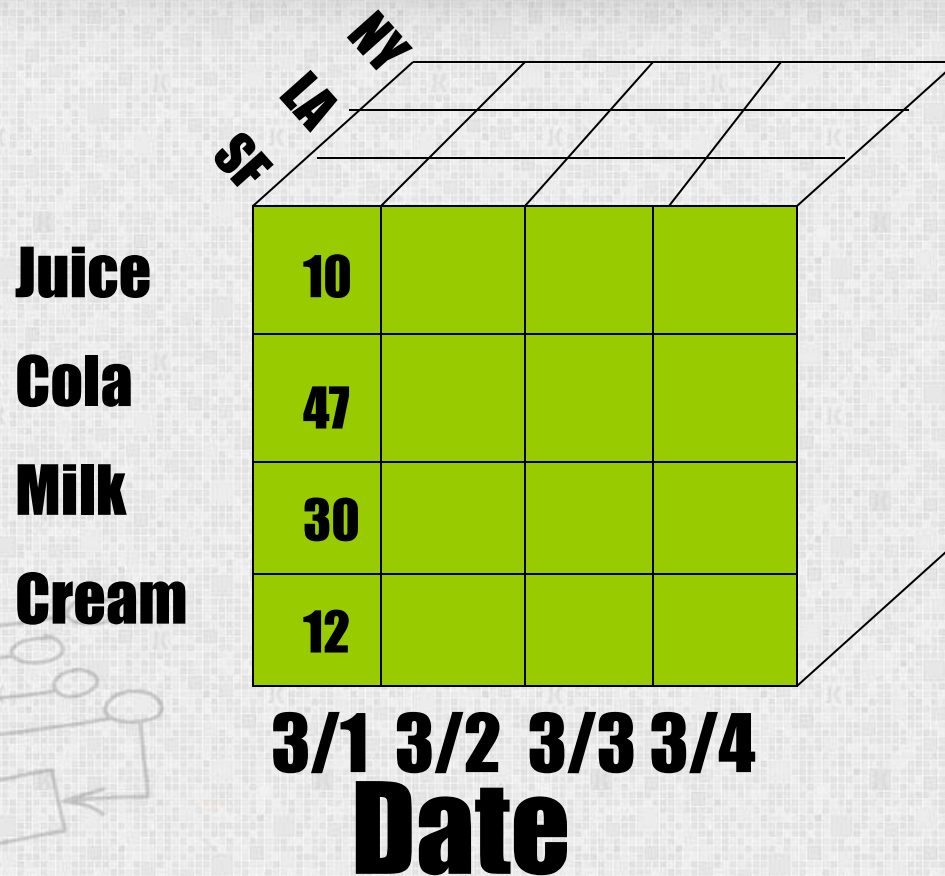


Online Analytical Processing (OLAP) Tools

- The use of a set of graphical tools that provides users with multidimensional views of their data and allows them to analyze the data



Multidimensional Data

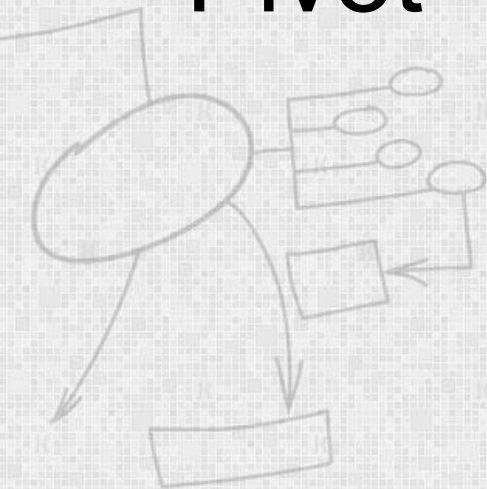


**Sales
Volume
as a
function
of time,
city and
product**

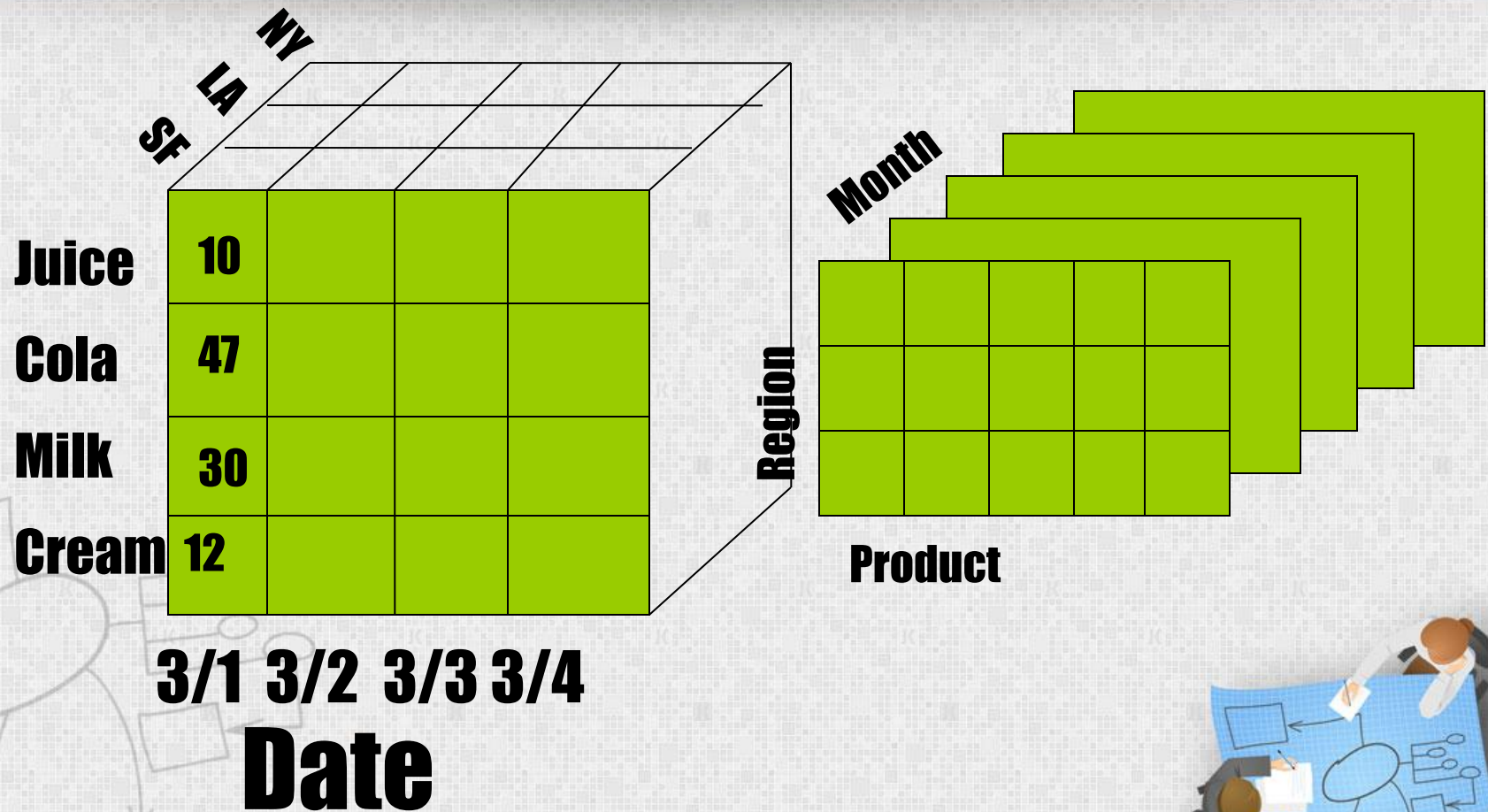


Operations in Multidimensional Data Model

- Drill Down
- Roll Up
- Slice and Dice
- Pivot



A Visual Operation: Pivot (Rotate)



Example of drill-down

Starting with summary data, users can obtain details for particular cells

a) Summary report

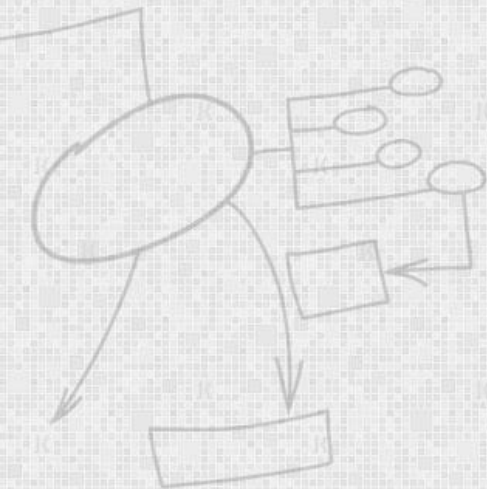
Brand	Package size	Sales
SofTowel	2-pack	\$75
SofTowel	3-pack	\$100
SofTowel	6-pack	\$50

b) Drill-down with color attribute added

Brand	Package size	Color	Sales
SofTowel	2-pack	White	\$30
SofTowel	2-pack	Yellow	\$25
SofTowel	2-pack	Pink	\$20
SofTowel	3-pack	White	\$50
SofTowel	3-pack	Green	\$25
SofTowel	3-pack	Yellow	\$25
SofTowel	6-pack	White	\$30
SofTowel	6-pack	Yellow	\$20

Data Mining

- Knowledge discovery using a blend of statistical, AI, and computer graphics techniques



Data Mining Techniques

- Regression (Linear, Logistic)
- Classification
- Clustering
- Outlier Detection

